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COMP4139
Machine Learning (MLE)

- Mathematics

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Learning Objectives

- Module overview continued
  - Gradient Descent
  - Probability Theory
- Data Representation Visualisation
- Curse of Dimensionality Concept

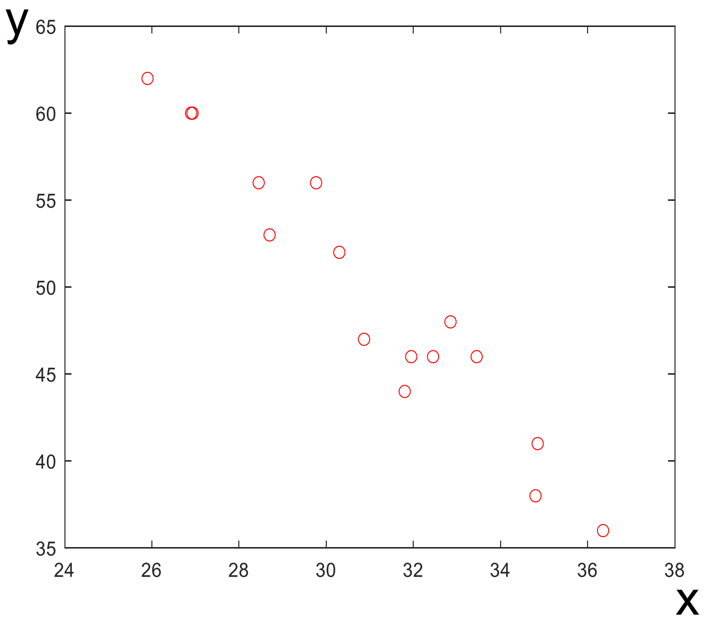
Gradient Descent Optimisation

Gradient descent is an iterative optimisation algorithm used to find a local minimum or maximum of a given function. This is one of the mostly used optimisation methods in machine learning.

Demo (Gradient Descent based Line Fitting)

Aim: to find parameter w*=[a*,b*] for a line that best fit these points (x, y)

a, b : parameter to be estimated k : iteration N : number of data samples α: learning rate



loss function: $f(a,b) = \sum_{n=1}^{N} \left(y_n - (ax_n + b)\right)^2$

Calculate partial derivatives:

$$\frac{\partial f}{\partial a^k} = -2 \sum_{n=1}^{N} \left(y_n - a^k x_n - b^k\right) \cdot x_n$$

$$\frac{\partial f}{\partial b^k} = -2 \sum_{n=1}^{N} \left(y_n - a^k x_n - b^k\right)$$

Initialise a, b, α when k=1

Iteratively updat a and b: $a^{k+1} = a^k - \alpha \frac{\partial f}{\partial a^k}$, $b^{k+1} = b^k - \alpha \frac{\partial f}{\partial b^k}$

Iterate until $f(a^k, b^k)$ is small enough or k>max iteration

How gradient descent works?

This is generalised to multiple variables rather than only one variable *w*.

Input: *f*: $R^n \longrightarrow R$ a differentiable function

$w^{(0)}$ an initial solution

Output: w^*, a local minimum of the cost function *f*.

begin

k $\leftarrow$ 0;

while *stop-crit* **and** (k< k_{max}) **do**

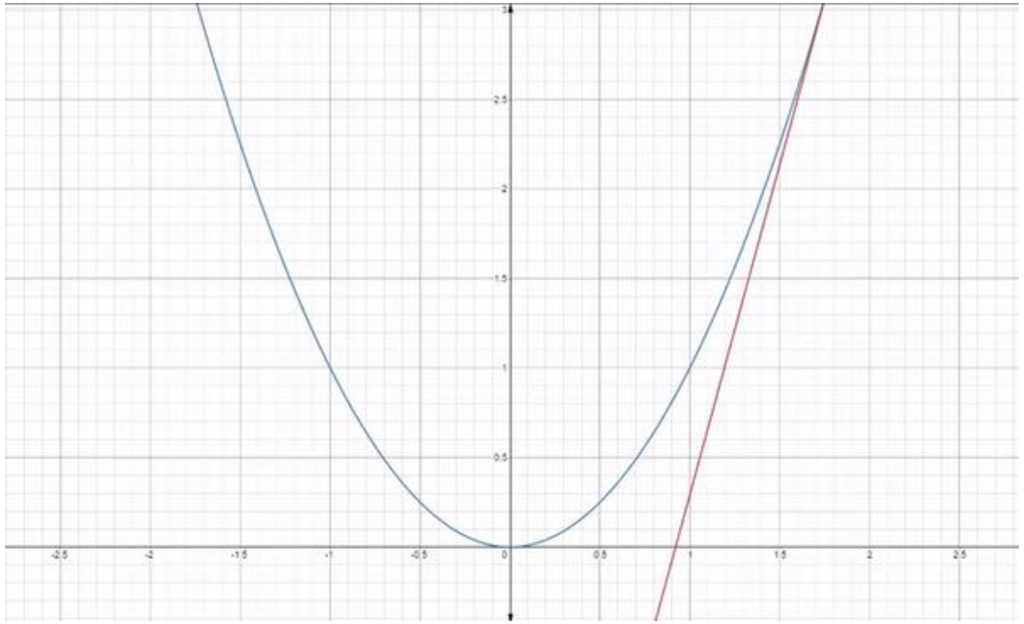
$w^{(k+1)} \leftarrow w^{(k)} - \alpha^{(k)} \nabla f(w)$;

with $\alpha^{(k)} = argmin_{\alpha \in R_+} f(w^{(k)} - \alpha \nabla f(w))$;

k $\leftarrow$ k+1;

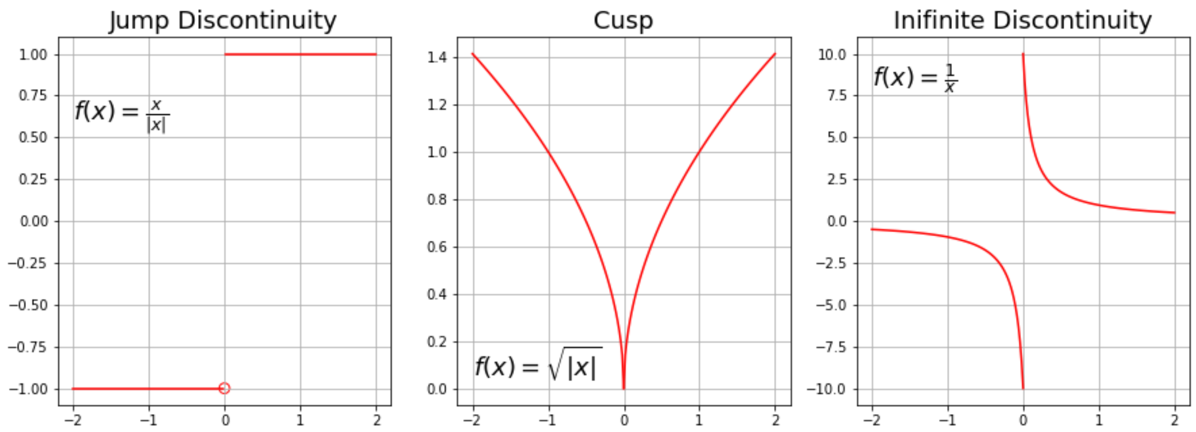
return $w^{(k)}$

end

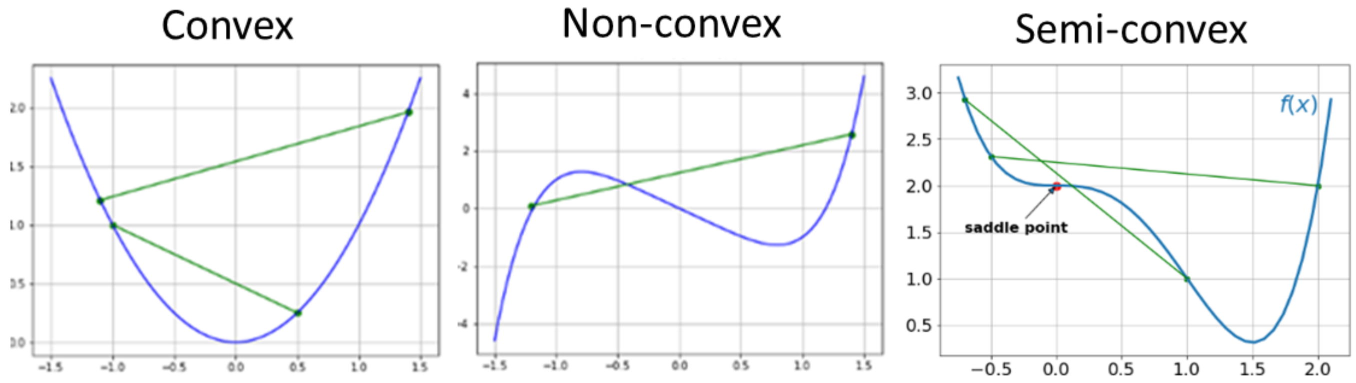


Requirements and Limitations of Gradient Descent

- The function to be optimised is differentiable at each point of *x* (no discontinuity).

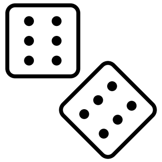


- The function is convex: the line segment connecting two function's points lay above the curve (does not cross it).



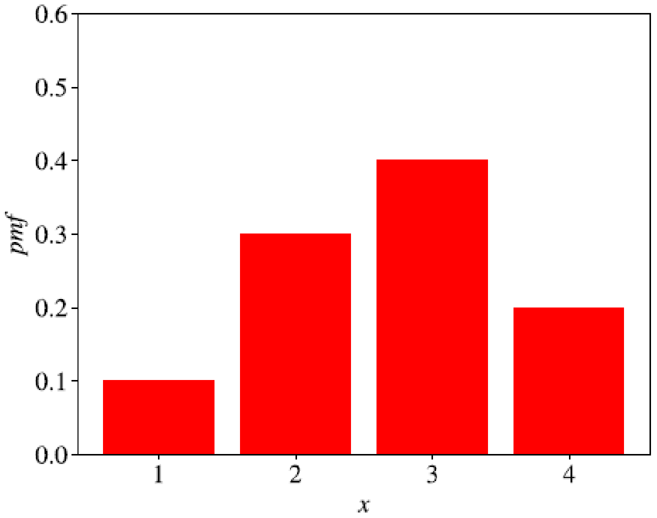
Probability- Random Variable (Discrete)

It is a variable *X* whose possible values are numerical outcomes of a random process. It can be discrete or continues.



- Discrete random variable is countable numbers or states.
- The probability distribution of a discrete random variable is a list of probabilities associated with each of its possible values (called probability mass function)
- Pr(*X*=1)=0.1, etc. The sum of all probabilities of all possible values is 1.
- Let *X* have *k* possible values $\{x_i\}_{i=1}^{k}$. The expectation E[x] is:

$$\mathbb{E}[X] \overset{\text{def}}{=} \sum_{i=1}^{k} x_i \Pr(X = x_i) = x_1 \Pr(X = x_1) + x_2 \Pr(X = x_2) + \cdots + x_k \Pr(X = x_k)$$



Probability- Random Variable (Continuous)

- Continuous random variable is an infinite number in a possible value range. E.g. time, height, weight, etc.

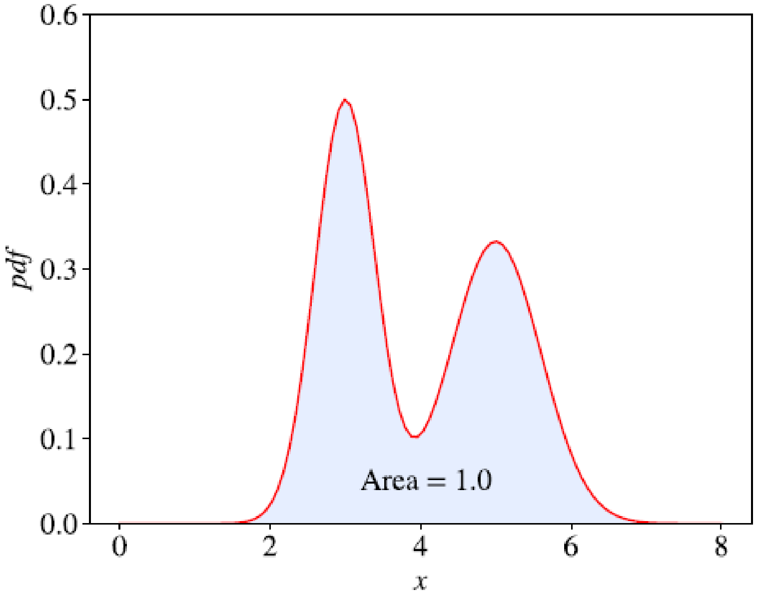
- The probability distribution of a continuous random variable is described by a probability density function.

- The area under the curve is 1.

- The expectation of a continuous random variable *X* is given by:

$$\mathbb{E}[X] \stackrel{\text{def}}{=} \int_{\mathbb{R}} x f_X(x)\, dx,$$

- We don't know *fx* in most cases in ML. We estimate it by collecting samples.



Bayes' Rule

- In many practical problems, the probability of one event occurs is often conditioned on another event. Taking account of priors!
- The conditional probability *Pr(X=x|Y=y)* is the probability of the random variable *X* to have a specific value *x* given that another random variable *Y* has a specific value of *y*.
- The Bayes' Rule is:

$$\Pr(X = x|Y = y) = \frac{\Pr(Y = y|X = x)\Pr(X = x)}{\Pr(Y = y)}$$

Event A and B

https://www.youtube.com/watch?v=HZGCoVF3YvM (nice video of only 15mins)

- Proof:

Probability of both A and B are true

Based on conditional probability, we have $P(A \mid B) = \frac{P(A \cap B)}{P(B)}, \text{ if } P(B) \neq 0$

Similarly, we have $P(B \mid A) = \frac{P(A \cap B)}{P(A)}, \text{ if } P(A) \neq 0,$

Combining the above, we have $P(A \mid B) = \frac{P(B \mid A)P(A)}{P(B)}, \text{ if } P(B) \neq 0.$

Example

$$P(\omega)$$

X	ω
0.2	ω_1
0.3	ω_1
0.6	ω_2
0.4	ω_2
0.6	ω_1
0.2	ω_1
0.8	ω_2
0.5	ω_1

What are P(w1), P(w2), P(x=0.2) ?

JOINT PROBABILITY DISTRIBUTION

data:

X	ω
0.2	ω_1
0.3	ω_1
0.6	ω_2
0.4	ω_2
0.6	ω_1
0.2	ω_1
0.8	ω_2
0.5	ω_1
0.2	ω_1
0.6	ω_2

Assuming features are discrete variables:

X	ω	$P(\omega_i, x_j)$
0.2	ω_1	0.3
0.3	ω_1	0.1
0.4	ω_2	0.1
0.5	ω_1	0.1
0.6	ω_1	0.1
0.6	ω_2	0.2
0.8	ω_2	0.1

All other combinations have 0 probability

Bayes' Theorem

- Note that the joint probability can be factorised in two ways:

$$P(\omega_j, x) = p(\omega_j|x)p(x) = p(x|\omega_j)P(\omega_j)$$

- This leads to Bayes' theorem, which calculates the class posterior probability in terms of the likelihood, prior, and evidence:

$$P(\omega_j|x) = \frac{p(x|\omega_j)P(\omega_j)}{p(x)}$$

Use Bayes' theorem to predict

$$P(\omega)$$

X	ω
0.2	ω_1
0.3	ω_1
0.6	ω_2
0.4	ω_2
0.6	ω_1
0.2	ω_1
0.8	ω_2
0.5	ω_1

$$P(\omega_j|x) = \frac{p(x|\omega_j)P(\omega_j)}{p(x)}$$

What is $P(w_1|x=0.2)$?

Frequentist vs Bayesian

- Frequentist:
  - Long-term frequency of an event occurring.
  - Data driven.

- Bayesian:
  - Assumes a prior model (knowledge) and update the model using data.
  - Model driven.